Pericom Series
Schematic Check List

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Revision History

Date	Revision #	Description
07/20/2020	1.0	1. Release
09/09/2020	1.1	1. Change the series AC couple capacitor of Reference Clock Input Pairs to DC couple (0ohm) for PI7C9X3G1632GP / PI7C9X3G808GP
12/09/2020	1.2	1. Correct GPIO[30] description for PI7C9X3G1632GP / PI7C9X3G808GP
02/09/2021	1.3	1. Change the pin name as HP_PDC_L[x] to PDC_L[x] for PI7C9X3G1632GP / PI7C9X3G808GP / PI7C9X3G606GP
03/23/2021	1.4	1. Add PI7C9X3G816GP part description 2. Correct EECK/ Debug_Mode_EN_L description for PI7C9X3G1632GP / PI7C9X3G808GP / PI7C9X3G606GP
06/29/2021	1.5	1. Update an notice at all differential signal and PDC_L[X] for PI7C9X3G1632GP / PI7C9X3G808GP / PI7C9X3G606GP / PI7C9X3G816GP 2. Add check items for all serial products. 3. Add CDEP Port configuration check for all serial products.
11/12/2021	1.6	1. Correct EECK/ Debug_Mode_EN_L description for PI7C9X3G1632GP / PI7C9X3G808GP / PI7C9X3G606GP
02/21/2022	1.7	1. Add TEST pin description for PI7C9X3G816GP
04/12/2022	1.8	1. Correct EECK frequency from 1.56MHz to 1.58MHz for PI7C9X3G1632GP / PI7C9X3G808GP / PI7C9X3G606GP / PI7C9X3G816GP
09/22/2022	1.9	1. Add PI7C9X3G1224GP part description

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1. Introduction

This application note is a schematic review check list for Pericom Series product. It gives requirements concerning the different pin connections that must be considered before starting any new board design. It does not consider PCB layout constraints.

This application note is not intended to be exhaustive. Its objective is to cover as many configurations of use as possible.

2. Associated Documentation

The Design Kit includes all documents you need. Please ask it from Pericom Distributor.

3. Schematic Check List

3.1. For GENIII PI7C9X3G1632GP

Section	Remark	Done
Block Diagram	● Customer should provide the block diagram to check	□
Port Configuration	● What port configuration in customer`s circuit.	□
Internal / External Clock Design In Circuit	● What kind of the clock source in customer`s circuit.	□
Reference Clock Source Type	● What kind of the clock type in customer`s circuit.	□
Parallel / Serial Hot Plug function implement (if chip supported)	● Does customer implement the hot plug function in circuit?	□
PM L1.1 function implement (if chip supported)	● Does customer implement the PM L1.1 function in circuit?	□
Fan out mode(Normal operate) or CDEP(Cross-Domain End-Device) Mode	● What kind of operation mode in customer`s circuit.	□

Section	Signal Name	Check Item	Remark	Done
Clock Source	REFCLKP_0[3:0] REFCLKN_0[3:0] REFCLKP_1[3:0] REFCLKN_1[3:0]	Connect to external 100MHz differential clock	1. The reference clock input is an unterminated AC coupled input. So the off-chip clock source must be terminated with both serial and parallel resistor network. Please refer to PCIe CEM specification for detail on how to realize on-board termination implementation. 2. When connecting the wires, please pay attention and check if the polarity and netnames are correct between your wires and the pins on the other side.	□
Reference Clock Input Pairs	REFCLKIP_0 REFCLKIN_0 REFCLKIP_1 REFCLKIN_1	Connect to external 100MHz differential clocks for the integrated reference clock buffers in two groups	1. If the reference clock input is HCSL type, it can be DC coupled. 2. If the 100MHz clock source is not in HCSL protocol (ex: LVPECL, LVDS, etc), please refer to application note to add application circuit. 3. When connecting the wires, please pay attention and check if the	□

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Section	Signal Name	Check Item	Remark	Done
			polarity and netnames are correct between your wires and the pins on the other side.	
Reference Clock Output Pairs	REFCLKOP_0[7:0] REFCLKON_0[7:0] REFCLKOP_1[7:0] REFCLKON_1[7:0]	100MHz external differential HCSL clock outputs from integrated reference clock buffer in six groups	1. Group 0 is REFCLKOP_0[7:0] and REFCLKON_0[7:0] 2. Group 1 is REFCLKOP_1[7:0] and REFCLKON_1[7:0] 3. Each group has 8 fanouts for all of the ports in one tile 4. When connecting the wires, please pay attention and check if the polarity and netnames are correct between your wires and the pins on the other side.	□
Reference Clock Power Down	CKBUFPD0_L CKBUFPD1_L	Integrated Reference Clock Buffer Power Down Signals: These signals CKBUFPDx_L, where x=0 and 1, are used to shut down the corresponding clock buffer in tile x. When CKBUFPDx_L is asserted low, the integrated reference clock buffer and reference clock outputs are disabled.		
Transmit and Receive Differential Pairs	PETP_0[15:0] PETN_0[15:0] PETP_1[15:0] PETN_1[15:0] PERP_0[15:0] PERN_0[15:0] PERP_1[15:0] PERN_1[15:0]	AC coupling capacitors of 220 nF should be placed at the same location (as close as possible) and should not be staggered from one trace to the other within the pair	1. This applies to all differential pairs of each port 2. It is fine to open un-used lane. 3. When connecting the wires, please pay attention and check if the polarity and netnames are correct between your wires and the pins on the other side.	□
RESET	PERST_L	Pull-up and connect to the system reset		□
External Reference Resistor	RESREF_0 RESREF_1	Each tile has its own reference resistor connection signal. Attach RESREF_x, where x=0 and 1, an external resistor with the precision of 200 ohm 1% 100-ppm/C to ground on the board. The reference resistor is used for calibration of RX and TX termination when the chip comes out of reset or a manual PHY tuning request is made		□
EEPROM	EECK EEDI EEDO	SPI Bus Serial EEPROM	1. Suggest reserving eeprom circuit for safety. 2. AT25XE011 or compatible SPI	□

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Section	Signal Name	Check Item	Remark	Done
	EECS_L		EEPROM parts. 3. EECK frequency is 1.58M. Please make sure selected SPI eeprom can support.	
JTAG	TCK TMS TDO_[1:0] TDI TRST_L JTAG_SEL_L	TMS and TRST_L Signals should be pulled low through a 330 ohm pull-down resistor if JTAG boundary scan function is not implemented JTAG_SEL_L for switch JTAG and PHY CR parallel register interface control.	1. Pull-down through 330 ohm at TMS and TRST_L for normal operation. 2. Pull-up through 5.1K ohm at JTAG_SEL_L for normal operation	□
Serial Hot-Plug	SHCL_I2C_0 SHCL_I2C_1	It is connected to SCL pin of all I2C IO expanders	1. I/O expander configure serial hot-plug port as follow Example 1. Set 011 at A[2:0] of I/O expander when SHP port configured as Port 4 of 3G1632GP. Example 2. Set 100 at A[2:0] of I/O expander when SHP port configured as Port 5 of 3G1632GP. 2. Plug high this pin through 5.1K ohm resistor if do not implement serial hot plug function.	□
	SHDA_I2C_0 SHDA_I2C_1	It is connected to SDA pin of all I2C IO expanders		□
	SHPCINT0_L SHPCINT1_L	It is connected to INT# output pin of all I2C IO expanders		□
	GPIO0[31]/SURPRISE_HP_EN	Disable Surprise Hot Plug Function: This signal is used to disable surprise hot function if SURPRISE_HP=0.	1. Pull-down through 330 ohm at SURPRISE_HP_EN for operational type is “managed”. 2. Pull-up through 5.1K ohm at SURPRISE_HP_EN for operational type is “surprised”.	
SMBUS /I2C	SDA_I2C	Bi-Directional System management or I2C Bus Data. This pin requires an external pull-up resistor.	NC if do not implement SMBUS/I2C function	□
	SCL_I2C	Bi-Directional System management or I2C Bus Clock. This pin requires an external pull-up resistor.	NC if do not implement SMBUS/I2C function	
	I2C_ADDR[2:0]	These pins are used to configure the value of the three least significant bits of the switch 7-bit Slave address. These pins require the external 5.1K-ohm pull-up resistors or 330-ohm pull-down resistors.	These pins require the external 5.1K ohm pull-up resistors or 330 ohm pull-down resistors.	□
	EECK/	EEPROM Clock: Clock signal	Pull-up through 5.1K ohm at EECK/	□

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Section	Signal Name	Check Item	Remark	Done
	Debug_Mode_EN_L	to 4-wire EEPROM interface. Debug Mode Enable (Debug_Mode_EN_L): During system initialization, EECK acts as the Debug_Mode_EN_L pin. In debug mode, it need be tired to low through a 330-ohm pull-down resistor. This pin has internal pull-up resistor. If no board trace is connected to this pin, the internal pull-up resistor of this pin is enough. However, if pin is connected to a board trace and not driven, it is recommended that an external 5.1K-ohm pull-up resistor be used.	Debug_Mode_EN_L for normal operation	
Misc Signals	GPIO[30]/ PHY_SRAM_ BYPASS	PHY_SRAM_BYPASS: when set, it bypasses PHY SRAM. This pin has internal pull-up. If no board trace is connected to this pin, the internal pull-up resistor of this pin is enough. However, if pin is connected to a board trace and not driven, it is recommended that an external 5.1K- ohm pull-up resistor be used.	Pull-up through 5.1K ohm at GPIO[30]/ PHY_SRAM_BYPASS for normal operation	□
	FATAL_ERR0_L/HOT_PLUG_EN0_L FATAL_ERR1_L/HOT_PLUG_EN1_L	Hot Plug Function Enable: Each switch tile has its own hot plug function enable signal to determine the downstream port is capable of handling either managed or surprised hot plug events. Besides, the GPIO pins would be redefined for hot-plug function if HOT_PLUG_ENx_L = 0. In total, there are 2 switch tiles and the corresponding hot plug enable signals are HOT_PLUG_EN_x, where x=0 and 1.	1. Pull-up through 5.1K ohm at this pin for disable hot plug function 2. Pull-down through 330 ohm at this pin for enable hot plug function	□
	INTA0_L/PM_L11_EN0_L INTA1_L/PM_L11_EN1_L	PM L1.1 Function Enable: Each switch tile has its own PM L1.1 function enable signal to determine the	1. Pull-up through 5.1K ohm at this pin for disable PM L1.1 function 2. Pull-down through 330 ohm at this pin for enable PM L1.1 function	□

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Section	Signal Name	Check Item	Remark	Done
		downstream port is capable of PM L1.1 function. Besides, the GPIO_x [15:8] pins would be redefined for PM L1.1 function if PM_L11_ENx_L = 0. In total, there are 2 switch tiles and the corresponding PM L1.1 function enable signals are PM_L11_ENx_L, where x=0 and 1.		
	CLKBUF_CMOS_EN0_L CLKBUF_CMOS_EN1_L	Intneral Reference Clock Input Select: Used to select the input for internal reference clock buffer is came from HCSL or CMOS. When CLKBUF_CMOS_ENx_L is asserted, CMOS is selected.	Pull-up through 5.1K ohm at both CLKBUF_CMOS_EN0_L and CLKBUF_CMOS_EN1_L for normal operation	□
	CHIPMODE[1:0]	Chip Operational Mode: These two input signals decide at which operational mode the chip is chosen	Pull-down through 330 ohm at CHIPMODE[1:0] for normal operation	□
	CKMODE_0 CKMODE_1	Clock Operational Mode: Each tile has its own clock operation mode signal (CKMODE_0 for tile #0 while CKMODE_1 for tile #1), which defines the relationship between physical SERDES lanes and reference clock. When CKMODE is 0, all 16 lanes (0 ~15) of SERDES are driven by one reference clock source. When CKMODE is 1, 16 lanes of SERDES are driven by two separate reference clock source via three pairs of reference clock inputs.	1. Pull-up through 5.1K ohm at this pin for CKMODE=1 function, which REFCLKP/N [3:0] pairs need to connect from external clock source. 2. Pull-down through 330 ohm at this pin for CKMODE=0 function, which REFCLKP/N [3:1] pairs are unused, and required pull-down 330 ohm at REFCLKP/N [3:1] pin. External clock source only connect to REFCLKP/N [0] pair.	□
	SWP_MODE[1:0]	Switch Partition Mode: The switch can be partitioned into multiple switches based upon the value of SWP_MODE[1:0].	Pull-down both of SWP_MODE[1:0] pin for normal operate.	□
	PDC0_L[7:0] PDC1_L[7:0]	Present Detect: When PDC_L is asserted low, it indicates this port is present. Otherwise, it indicates this port is absent.	1. Pull-down if chip to chip connection. 2. Pull-up and connect to pin PRSNT of slot if chip to slot connection. 3. NC if unused. 4. When connecting the wires, please pay attention and check if	□

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Section	Signal Name	Check Item	Remark	Done
			the netnames are correct between your wires and the pins on the other side.	
Port Specific	PORTCFG_0[2:0] PORTCFG_1[2:0]	Port Configuration: Each switch tile has its own port configuration, which determines how 16 lanes are distributed among ports. In total, there are two switch tiles and the corresponding port configuration signals are PORTCFG_x[2:0], where x=0 and 1.	1. By default, the upstream port is port 0 2. PORTCFG[2:0] as 000b, 1Port-16Lane Configuration. 3. PORTCFG[2:0] as 001b, 2Port-16Lane Configuration. 4. PORTCFG[2:0] as 010b, 3Port-16Lane Configuration. 5. PORTCFG[2:0] as 011b, 4Port-16Lane Configuration. 6. PORTCFG[2:0] as 100b, 5Port-16Lane Configuration. 7. PORTCFG[2:0] as 101b, 8Port-16Lane Configuration. These pins require the external 5.1K-ohm pull-up resistors or 330-ohm pull-down resistors.	
Power Supply and Signal Ground	VDDC	0.95V power. Used as digital core power pins.	1. Decoupling/Filtering capacitors must be added. 2. HI1812V101R-10 bead part is required at 0.95V power rail which VDDC/C_VDDC/VP.	□
	VDDR	1.8V power. Used as digital I/O power pins.	1. Decoupling/Filtering capacitors must be added. 2. HI1812V101R-10 bead part is required at 0.95V power rail which VDDC/C_VDDC/VP.	□
	C_VDDR	1.8V power. Used as reference clock power pins.		□
	C_VDDC	0.95V power. Used as reference clock power pins.		□
	VPH	1.8V power. Used as PCI Express analog high voltage power pins.		□
	VP	0.95V power. Used as PCI Express analog and core power pins.		□
	VSS	Used as ground pins.		□

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3.2. For GENIII PI7C9X3G808GP

Section	Remark	Done
Block Diagram	● Customer should provide the block diagram to check	□
Port Configuration	● What port configuration in customer`s circuit.	□
Internal / External Clock Design In Circuit	● What kind of the clock source in customer`s circuit.	□
Reference Clock Source Type	● What kind of the clock type in customer`s circuit.	□
Parallel / Serial Hot Plug function implement (if chip supported)	● Does customer implement the hot plug function in circuit?	□
PM L1.1 function implement (if chip supported)	● Does customer implement the PM L1.1 function in circuit?	□
Fan out mode(Normal operate) or CDEP(Cross-Domain End-Device) Mode	● What kind of operation mode in customer`s circuit.	□

Section	Signal Name	Check Item	Remark	Done
Clock Source	REFCLKP[1:0] REFCLKN[1:0]	Connect to external 100MHz differential clock	1. The reference clock input is an unterminated AC coupled input. So the off-chip clock source must be terminated with both serial and parallel resistor network. Please refer to PCIe CEM specification for detail on how to realize on-board termination implementation. 2. When connecting the wires, please pay attention and check if the polarity and netnames are correct between your wires and the pins on the other side.	□
Reference Clock Input Pairs	REFCLKIP REFCLKIN	Connect to external 100MHz differential clocks for the integrated reference clock buffer	1. If the reference clock input is HCSL type, it can be DC coupled. 2. If the 100MHz clock source is not in HCSL protocol (ex: LVPECL, LVDS, etc), please refer to application note to add application circuit.	□
Reference Clock Output Pairs	REFCLKOP[7:0] REFCLKON[7:0]	100MHz external differential HCSL clock outputs from integrated reference clock buffer.	1. When connecting the wires, please pay attention and check if the polarity and netnames are correct between your wires and the pins on the other side.	□
Reference Clock Power Down	CKBUFPD_L	Integrated Reference Clock Buffer Power Down Signal: This signal CKBUFPD_L is used to shut down the integrated clock buffer. When CKBUFPD_L is asserted low, the integrated reference clock buffer and reference clock outputs are disabled.		

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Section	Signal Name	Check Item	Remark	Done
Transmit and Receive Differential Pairs	PETP[7:0] PETN[7:0] PERP[7:0] PERN[7:0]	AC coupling capacitors of 220 nF should be placed at the same location (as close as possible) and should not be staggered from one trace to the other within the pair	1. This applies to all differential pairs of each port 2. It is fine to open un-used lane. 3. When connecting the wires, please pay attention and check if the polarity and netnames are correct between your wires and the pins on the other side.	□
RESET	PERST_L	Pull-up and connect to the system reset		□
External Reference Resistor	RESREF	Reference Resistor Connection: Attach RESREF an external resistor with the precision of 200 ohm 1% 100-ppm/C to ground on the board. The reference resistor is used for calibration of RX and TX termination when the chip comes out of reset or a manual PHY tuning request is made.		□
EEPROM	EECK EEDI EEDO EECS_L	SPI Bus Serial EEPROM	1. Suggest to reserve eeprom circuit for safety. 2. AT25XE011 or compatible SPI EEPROM parts. 3. EECK frequency is 1.58M. Please make sure selected SPI eeprom can support.	□
JTAG	TCK TMS TDO TDI TRST_L JTAG_SEL_L	TMS and TRST_L Signals should be pulled low through a 330 ohm pull-down resistor if JTAG boundary scan function is not implemented JTAG_SEL_L for switch JTAG and PHY CR parallel register interface control.	1. Pull-down through 330 ohm at TMS and TRST_L for normal operation. 2. Pull-up through 5.1K ohm at JTAG_SEL_L for normal operation	□
Serial Hot-Plug	SHCL_I2C	It is connected to SCL pin of all I2C IO expanders	1. I/O expander configure serial hot-plug port as follow Example 1. Set 011 at A[2:0] of I/O expander when SHP port configured as Port 4 of 3G808GP. Example 2. Set 100 at A[2:0] of I/O expander when SHP port configured as Port 5 of 3G808GP. 2. Plug high this pin through 5.1K ohm resistor if do not implement serial hot plug function.	□
	SHDA_I2C	It is connected to SDA pin of all I2C IO expanders		□
	SHPCINT_L	It is connected to INT# output pin of all I2C IO expanders		□
	GPIO0[31]/SURPRI	Disable Surprise Hot Plug	1. Pull-down through 330 ohm at	

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Section	Signal Name	Check Item	Remark	Done
	SE_HP	Function: This signal is used to disable surprise hot function if SURPRISE_HP=0.	SURPRISE_HP_EN for operational type is “managed”. 2. Pull-up through 5.1K ohm at SURPRISE_HP_EN for operational type is “surprised”.	
SMBUS /I2C	SDA_I2C	Bi-Directional System management or I2C Bus Data. This pin requires an external pull-up resistor.	NC if do not implement SMBUS/I2C function	□
	SCL_I2C	Bi-Directional System management or I2C Bus Clock. This pin requires an external pull-up resistor.	NC if do not implement SMBUS/I2C function	
	I2C_ADDR[2:0]	These pins are used to configure the value of the three least significant bits of the switch 7-bit Slave address. These pins require the external 5.1K-ohm pull-up resistors or 330-ohm pull-down resistors.	These pins require the external 5.1K ohm pull-up resistors or 330 ohm pull-down resistors.	□
	EECK/ Debug_Mode_EN_L	EEPROM Clock: Clock signal to 4-wire EEPROM interface. Debug Mode Enable (Debug_Mode_EN_L): During system initialization, EECK acts as the Debug_Mode_EN_L pin. In debug mode, it need be tired to low through a 330-ohm pull-down resistor. This pin has internal pull-up resistor. If no board trace is connected to this pin, the internal pull-up resistor of this pin is enough. However, if pin is connected to a board trace and not driven, it is recommended that an external 5.1K-ohm pull-up resistor be used.	Pull-up through 5.1K ohm at EECK/ Debug_Mode_EN_L for normal operation	□
Misc Signals	GPIO[30]/ PHY_SRAM_ BYPASS	PHY_SRAM_BYPASS: when set, it bypasses PHY SRAM. This pin has internal pull-up. If no board trace is connected to this pin, the internal pull-up resistor of this pin is enough. However, if pin is connected to a board trace and not driven, it is recommended that an	Pull-up through 5.1K ohm at GPIO[30]/ PHY_SRAM_BYPASS for normal operation	□

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Section	Signal Name	Check Item	Remark	Done
		external 5.1K- ohm pull-up resistor be used.		
	FATAL_ERR_L/ HOT_PLUG_EN_L	Hot Plug Function Enable: It is used to determine the downstream port is capable of handling either managed or surprised hot plug events. Besides, the GPIO pins would be redefined for hot-plug function if HOT_PLUG_EN_L = 0.	1. Pull-up through 5.1K ohm at this pin for disable hot plug function 2. Pull-down through 330 ohm at this pin for enable hot plug function	□
	INTA_L/PM_L11_EN_L	PM L1.1 Function Enable: It is used to determine the downstream port is capable of PM L1.1 function. Besides, the GPIO[15:8] pins would be redefined for PM L1.1 function if PM_L11_EN_L = 0.	1. Pull-up through 5.1K ohm at this pin for disable PM L1.1 function 2. Pull-down through 330 ohm at this pin for enable PM L1.1 function	□
	CLKBUF_INPUT_SEL	I Clock Buffer Input Select: It is used to select the input of integrated ref clock buffer. When set high, the input is came the external reference clock source through REFCLKIP/N pin. When set low, the input is came from internal PHY clock and REFCLKIP/N pin need be tied to GND through 330-ohm resistor.	Pull-up through 5.1K ohm at CLKBUF_INPUT_SEL pin for normal operation	□
	CHIPMODE[1:0]	Chip Operational Mode: These two input signals decide at which operational mode the chip is chosen	Pull-down through 330 ohm at CHIPMODE[1:0] for normal operation	□
	CKMODE	Clock Operational Mode: It is used define the relationship between physical SERDES lanes and reference clock. When CKMODE is 0, all 8 lanes (0 ~7) of SERDES are driven by one reference clock source. When CKMODE is 1, 8 lanes of SERDES are driven by two separate reference clock source via two pairs of reference clock inputs.	1. Pull-up through 5.1K ohm at this pin for CKMODE=1 function, which REFCLKP/N [1:0] pairs need to connect from external clock source. 2. Pull-down through 330 ohm at this pin for CKMODE=0 function, which REFCLKP/N [1] pairs are unused, and required pull-down 330 ohm at REFCLKP/N [1] pin. External clock source only connect to REFCLKP/N [0] pair.	□
	PDC_L[7:0]	Present Detect: When PDC_L is asserted low, it indicates this port is present. Otherwise, it indicates this port is absent.	1. Pull-down if chip to chip connection. 2. Pull-up and connect to pin PRSNT of slot if chip to slot connection.	□

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Section	Signal Name	Check Item	Remark	Done
			3. NC if unused. 4. When connecting the wires, please pay attention and check if the netnames are correct between your wires and the pins on the other side.	
Port Specific	PORTCFG[2:0]	Port Configuration: They are used to determine how 8 lanes are distributed among ports.	1. By default, the upstream port is port 0 2. PORTCFG[2:0] as 001b, 2Port-8Lane Configuration. 3. PORTCFG[2:0] as 010b, 3Port-8Lane Configuration. 4. PORTCFG[2:0] as 011b, 4Port-8Lane Configuration. 5. PORTCFG[2:0] as 100b, 5Port-8Lane Configuration. 6. PORTCFG[2:0] as 101b, 8Port-8Lane Configuration. These pins require the external 5.1K-ohm pull-up resistors or 330-ohm pull-down resistors.	
Power Supply and Signal Ground	VDDC	0.95V power. Used as digital core power pins.	1. Decoupling/Filtering capacitors must be added. 2. HI1812V101R-10 bead part is required at 0.95V power rail which VDDC/C_VDDC/VP.	□
	VDDR	1.8V power. Used as digital I/O power pins.	1. Decoupling/Filtering capacitors must be added. 2. HI1812V101R-10 bead part is required at 0.95V power rail which VDDC/C_VDDC/VP.	□
	C_VDDR	1.8V power. Used as reference clock power pins.		□
	C_VDDC	0.95V power. Used as reference clock power pins.		□
	VPH	1.8V power. Used as PCI Express analog high voltage power pins.		□
	VP	0.95V power. Used as PCI Express analog and core power pins.		□
	VSS	Used as ground pins.		□

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3.3. For GENIII PI7C9X3G606GP

Section	Remark	Done
Block Diagram	● Customer should provide the block diagram to check	□
Port Configuration	● What port configuration in customer`s circuit.	□
Internal / External Clock Design In Circuit	● What kind of the clock source in customer`s circuit.	□
Reference Clock Source Type	● What kind of the clock type in customer`s circuit.	□
Parallel / Serial Hot Plug function implement (if chip supported)	● Does customer implement the hot plug function in circuit?	□
PM L1.1 function implement (if chip supported)	● Does customer implement the PM L1.1 function in circuit?	□
Fan out mode(Normal operate) or CDEP(Cross-Domain End-Device) Mode	● What kind of operation mode in customer`s circuit.	□

Section	Signal Name	Check Item	Remark	Done
Clock Source	REFCLKP[1:0] REFCLKN[1:0]	Connect to external 100MHz differential clock	1. The reference clock input is an unterminated AC coupled input. So the off-chip clock source must be terminated with both serial and parallel resistor network. Please refer to PCIe CEM specification for detail on how to realize on-board termination implementation. 2. When connecting the wires, please pay attention and check if the polarity and netnames are correct between your wires and the pins on the other side.	□
Reference Clock Output Pairs	REFCLKOP[7:4;1] REFCLKON[7:4;1]	100MHz external differential HCSL clock outputs from integrated reference clock buffer.	1. When connecting the wires, please pay attention and check if the polarity and netnames are correct between your wires and the pins on the other side.	□
Transmit and Receive Differential Pairs	PETP[5:0] PETN[5:0] PERP[5:0] PERN[5:0]	AC coupling capacitors of 220 nF should be placed at the same location (as close as possible) and should not be staggered from one trace to the other within the pair	1. This applies to all differential pairs of each port 2. It is fine to open un-used lane. 3. When connecting the wires, please pay attention and check if the polarity and netnames are correct between your wires and the pins on the other side.	□
RESET	PERST_L	Pull-up and connect to the system reset		□
External Reference Resistor	RESREF	Reference Resistor Connection: Attach RESREF an external resistor with the precision of 200 ohm 1% 100-ppm/C to ground on the board. The reference resistor		□

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Section	Signal Name	Check Item	Remark	Done
		is used for calibration of RX and TX termination when the chip comes out of reset or a manual PHY tuning request is made.		
EEPROM	EECK EEDI EEDO EECS_L	SPI Bus Serial EEPROM	1. Suggest to reserve eeprom circuit for safety. 2. AT25XE011 or compatible SPI EEPROM parts. 3. EECK frequency is 1.58M. Please make sure selected SPI eeprom can support.	□
JTAG	TCK TMS TDO TDI TRST_L JTAG_SEL_L	TMS and TRST_L Signals should be pulled low through a 330 ohm pull-down resistor if JTAG boundary scan function is not implemented JTAG_SEL_L for switch JTAG and PHY CR parallel register interface control.	1. Pull-down through 330 ohm at TMS and TRST_L for normal operation. 2. Pull-up through 5.1K ohm at JTAG_SEL_L for normal operation	□
Serial Hot-Plug	SHCL_I2C	It is connected to SCL pin of all I2C IO expanders	1. I/O expander configure serial hot-plug port as follow Example 1. Set 011 at A[2:0] of I/O expander when SHP port configured as Port 4 of 3G808GP. Example 2. Set 100 at A[2:0] of I/O expander when SHP port configured as Port 5 of 3G808GP. 2. Plug high this pin through 5.1K ohm resistor if do not implement serial hot plug function.	□
	SHDA_I2C	It is connected to SDA pin of all I2C IO expanders		□
	SHPCINT_L	It is connected to INT# output pin of all I2C IO expanders		□
SMBUS /I2C	SCL_I2C	Bi-Directional System management or I2C Bus Clock. This pin requires an external pull-up resistor.	NC if do not implement SMBUS/I2C function	□
	SDA_I2C	Bi-Directional System management or I2C Bus Data. This pin requires an external pull-up resistor.	NC if do not implement SMBUS/I2C function	□
Misc Signals	EECK/ Debug_Mode_EN_L	EEPROM Clock: Clock signal to 4-wire EEPROM interface. Debug Mode Enable (Debug_Mode_EN_L): During system initialization, EECK acts as the Debug_Mode_EN_L pin. In debug mode, it need be tired to low through a 330-ohm	Pull-up through 5.1K ohm at EECK/ Debug_Mode_EN_L for normal operation	□

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Section	Signal Name	Check Item	Remark	Done
		pull-down resistor. This pin has internal pull-up resistor. If no board trace is connected to this pin, the internal pull-up resistor of this pin is enough. However, if pin is connected to a board trace and not driven, it is recommended that an external 5.1K-ohm pull-up resistor be used.		
	FATAL_ERR_L/ HOT_PLUG_EN_L	Hot Plug Function Enable: It is used to determine the downstream port is capable of handling either managed or surprised hot plug events. Besides, the GPIO pins would be redefined for hot-plug function if HOT_PLUG_EN_L = 0.	1. Pull-up through 5.1K ohm at this pin for disable hot plug function 2. Pull-down through 330 ohm at this pin for enable hot plug function	□
	INTA_L/PM_L11_EN_L	PM L1.1 Function Enable: It is used to determine the downstream port is capable of PM L1.1 function. Besides, the GPIO[15:8] pins would be redefined for PM L1.1 function if PM_L11_EN_L = 0.	1. Pull-up through 5.1K ohm at this pin for disable PM L1.1 function 2. Pull-down through 330 ohm at this pin for enable PM L1.1 function	□
	CHIPMODE[1:0]	Chip Operational Mode: These two input signals decide at which operational mode the chip is chosen	Pull-down through 330 ohm at CHIPMODE[1:0] for normal operation	□
	CKMODE	Clock Operational Mode: It is used define the relationship between physical SERDES lanes and reference clock. When CKMODE is 0, all 8 lanes (0 ~7) of SERDES are driven by one reference clock source. When CKMODE is 1, 8 lanes of SERDES are driven by two separate reference clock source via two pairs of reference clock inputs.	1. Pull-up through 5.1K ohm at this pin for CKMODE=1 function, which REFCLKP/N [1:0] pairs need to connect from external clock source. 2. Pull-down through 330 ohm at this pin for CKMODE=0 function, which REFCLKP/N [1] pairs are unused, and required pull-down 330 ohm at REFCLKP/N [1] pin. External clock source only connect to REFCLKP/N [0] pair.	□
Misc Signals	TEST	Test: This pin should be tied to ground through a 330ohm pull-down resistor.	Pull-down through 330 ohm at TEST for normal operation	
	PDC_L[7:4;1]	Present Detect: When PDC_L is asserted low, it indicates this port is present. Otherwise, it indicates this port is absent.	1. Pull-down if chip to chip connection. 2. Pull-up and connect to pin PRSNT of slot if chip to slot connection. 3. NC if unused.	

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Section	Signal Name	Check Item	Remark	Done
			4. When connecting the wires, please pay attention and check if the netnames are correct between your wires and the pins on the other side.	
Port Specific	PORTCFG[2:0]	Port Configuration: They are used to determine how 6 lanes are distributed among ports.	1. By default, the upstream port is port 0 2. PORTCFG[2:0] as 010b, 3Port-6Lane Configuration. 3. PORTCFG[2:0] as 100b, 5Port-6Lane Configuration. 4. PORTCFG[2:0] as 101b, 6Port-6Lane Configuration. These pins require the external 5.1K-ohm pull-up resistors or 330-ohm pull-down resistors.	□
Power Supply and Signal Ground	VDDC	0.95V power. Used as digital core power pins.	1. Decoupling/Filtering capacitors must be added. 2. HI1812V101R-10 bead part is required at 0.95V power rail which VDDC/C_VDDC/VP.	□
	VDDR	1.8V power. Used as digital I/O power pins.		□
	C_VDDR	1.8V power. Used as reference clock power pins.		□
	C_VDDC	0.95V power. Used as reference clock power pins.		□
	VPH	1.8V power. Used as PCI Express analog high voltage power pins.		□
	VP	0.95V power. Used as PCI Express analog and core power pins.		□
	VSS	Used as ground pins.		□

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3.4. For GENIII PI7C9X3G816GP

Section	Remark	Done
Block Diagram	● Customer should provide the block diagram to check	□
Port Configuration	● What port configuration in customer`s circuit.	□
Internal / External Clock Design In Circuit	● What kind of the clock source in customer`s circuit.	□
Reference Clock Source Type	● What kind of the clock type in customer`s circuit.	□
Parallel / Serial Hot Plug function implement (if chip supported)	● Does customer implement the hot plug function in circuit?	□
PM L1.1 function implement (if chip supported)	● Does customer implement the PM L1.1 function in circuit?	□
Fan out mode(Normal operate) or CDEP(Cross-Domain End-Device) Mode	● What kind of operation mode in customer`s circuit.	□

Section	Signal Name	Check Item	Remark	Done
Clock Source	REFCLKP[2:0] REFCLKN[2:0]	Connect to external 100MHz differential clock	1. The reference clock input is an unterminated AC coupled input. So the off-chip clock source must be terminated with both serial and parallel resistor network. Please refer to PCIe CEM specification for detail on how to realize on-board termination implementation. 2. When connecting the wires, please pay attention and check if the polarity and netnames are correct between your wires and the pins on the other side.	□
Reference Clock Input Pairs	REFCLKIP REFCLKIN	Connect to external 100MHz differential clocks for the integrated reference clock buffer	1. If the reference clock input is HCSL type, it can be DC coupled. 2. If the 100MHz clock source is not in HCSL protocol (ex: LVPECL, LVDS, etc), please refer to application note to add application circuit.	□
Reference Clock Output Pairs	REFCLKOP[7:0] REFCLKON[7:0]	100MHz external differential HCSL clock outputs from integrated reference clock buffer.	1. When connecting the wires, please pay attention and check if the polarity and netnames are correct between your wires and the pins on the other side.	□
Reference Clock Power Down	CKBUFPD_L	Integrated Reference Clock Buffer Power Down Signal: This signal CKBUFPD_L is used to shut down the integrated clock buffer. When CKBUFPD_L is asserted low, the integrated reference clock buffer and reference clock outputs are disabled.		

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Section	Signal Name	Check Item	Remark	Done
Transmit and Receive Differential Pairs	PETP[15:0] PETN[15:0] PERP[15:0] PERN[15:0]	AC coupling capacitors of 220 nF should be placed at the same location (as close as possible) and should not be staggered from one trace to the other within the pair	1. This applies to all differential pairs of each port 2. It is fine to open un-used lane. 3. When connecting the wires, please pay attention and check if the polarity and netnames are correct between your wires and the pins on the other side.	□
RESET	PERST_L	Pull-up and connect to the system reset		□
External Reference Resistor	RESREF	Reference Resistor Connection: Attach RESREF an external resistor with the precision of 200 ohm 1% 100-ppm/C to ground on the board. The reference resistor is used for calibration of RX and TX termination when the chip comes out of reset or a manual PHY tuning request is made.		□
EEPROM	EECK EEDI EEDO EECS_L	SPI Bus Serial EEPROM	1. Suggest to reserve eeprom circuit for safety. 2. AT25XE011 or compatible SPI EEPROM parts. 3. EECK frequency is 1.58M. Please make sure selected SPI eeprom can support.	□
JTAG	TCK TMS TDO TDI TRST_L JTAG_SEL_L	TMS and TRST_L Signals should be pulled low through a 330 ohm pull-down resistor if JTAG boundary scan function is not implemented JTAG_SEL_L for switch JTAG and PHY CR parallel register interface control.	1. Pull-down through 330 ohm at TMS and TRST_L for normal operation. 2. Pull-up through 5.1K ohm at JTAG_SEL_L for normal operation	□
Serial Hot-Plug	SHCL_I2C	It is connected to SCL pin of all I2C IO expanders	1. I/O expander configure serial hot-plug port as follow Example 1. Set 011 at A[2:0] of I/O expander when SHP port configured as Port 4 of 3G816GP. Example 2. Set 100 at A[2:0] of I/O expander when SHP port configured as Port 5 of 3G816GP. 2. Plug high this pin through 5.1K ohm resistor if do not implement serial hot plug function.	□
	SHDA_I2C	It is connected to SDA pin of all I2C IO expanders		□
	SHPCINT_L	It is connected to INT# output pin of all I2C IO expanders		□
	GPIO0[31]/SURPRI	Disable Surprise Hot Plug	1. Pull-down through 330 ohm at	

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Section	Signal Name	Check Item	Remark	Done
	SE_HP	Function: This signal is used to disable surprise hot function if SURPRISE_HP=0.	SURPRISE_HP_EN for operational type is “managed”. 2. Pull-up through 5.1K ohm at SURPRISE_HP_EN for operational type is “surprised”.	
SMBUS /I2C	SDA_I2C	Bi-Directional System management or I2C Bus Data. This pin requires an external pull-up resistor.	NC if do not implement SMBUS/I2C function	□
	SCL_I2C	Bi-Directional System management or I2C Bus Clock. This pin requires an external pull-up resistor.	NC if do not implement SMBUS/I2C function	
	I2C_ADDR[2:0]	These pins are used to configure the value of the three least significant bits of the switch 7-bit Slave address. These pins require the external 5.1K-ohm pull-up resistors or 330-ohm pull-down resistors.	These pins require the external 5.1K ohm pull-up resistors or 330 ohm pull-down resistors.	□
	EECK/ Debug_Mode_EN_L	EEPROM Clock: Clock signal to 4-wire EEPROM interface. Debug Mode Enable (Debug_Mode_EN_L): During system initialization, EECK acts as the Debug_Mode_EN_L pin. In debug mode, it need be tired to low through a 330-ohm pull-down resistor. This pin has internal pull-up resistor. If no board trace is connected to this pin, the internal pull-up resistor of this pin is enough. However, if pin is connected to a board trace and not driven, it is recommended that an external 5.1K-ohm pull-up resistor be used.	Pull-up through 5.1K ohm at EECK/ Debug_Mode_EN_L for normal operation	□
Misc Signals	TEST	Test: This pin should be tied to ground through a 330ohm pull-down resistor.	Pull-down through 330 ohm at TEST for normal operation	□
	GPIO[30]/ PHY_SRAM_ BYPASS	PHY_SRAM_BYPASS: when set, it bypasses PHY SRAM. This pin has internal pull-up. If no board trace is connected to this pin, the internal pull-up resistor of this pin is enough. However,	Pull-up through 5.1K ohm at GPIO[30]/ PHY_SRAM_BYPASS for normal operation	□

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Section	Signal Name	Check Item	Remark	Done
		if pin is connected to a board trace and not driven, it is recommended that an external 5.1K- ohm pull-up resistor be used.		
	FATAL_ERR_L/ HOT_PLUG_EN_L	Hot Plug Function Enable: It is used to determine the downstream port is capable of handling either managed or surprised hot plug events. Besides, the GPIO pins would be redefined for hot-plug function if HOT_PLUG_EN_L = 0.	1. Pull-up through 5.1K ohm at this pin for disable hot plug function 2. Pull-down through 330 ohm at this pin for enable hot plug function	□
	INTA_L/PM_L11_EN_L	PM L1.1 Function Enable: It is used to determine the downstream port is capable of PM L1.1 function. Besides, the GPIO[15:8] pins would be redefined for PM L1.1 function if PM_L11_EN_L = 0.	1. Pull-up through 5.1K ohm at this pin for disable PM L1.1 function 2. Pull-down through 330 ohm at this pin for enable PM L1.1 function	□
	CLKBUF_CMOS_EN_L	I Clock Buffer Input Select: It is used to select the input of integrated ref clock buffer. When set high, the input is came the external reference clock source through REFCLKIP/N pin. When set low, the input is came from internal PHY clock and REFCLKIP/N pin need be tied to GND through 330-ohm resistor.	Pull-up through 5.1K ohm at CLKBUF_INPUT_SEL pin for normal operation	□
	CHIPMODE[1:0]	Chip Operational Mode: These two input signals decide at which operational mode the chip is chosen	Pull-down through 330 ohm at CHIPMODE[1:0] for normal operation	□
	CKMODE	Clock Operational Mode: It is used define the relationship between physical SERDES lanes and reference clock. When CKMODE is 0, all 16 lanes (0 ~15) of SERDES are driven by one reference clock source. When CKMODE is 1, 16 lanes of SERDES are driven by two separate reference clock source via two pairs of reference clock inputs.	1. Pull-up through 5.1K ohm at this pin for CKMODE=1 function, which REFCLKP/N [1:0] pairs need to connect from external clock source. 2. Pull-down through 330 ohm at this pin for CKMODE=0 function, which REFCLKP/N [1] pairs are unused, and required pull-down 330 ohm at REFCLKP/N [1] pin. External clock source only connect to REFCLKP/N [0] pair.	□
	PDC_L[7:0]	Present Detect: When PDC_L is asserted low, it	1. Pull-down if chip to chip connection.	□

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Section	Signal Name	Check Item	Remark	Done
		indicates this port is present. Otherwise, it indicates this port is absent.	2. Pull-up and connect to pin PRSNT of slot if chip to slot connection. 3. NC if unused. 4. When connecting the wires, please pay attention and check if the netnames are correct between your wires and the pins on the other side.	
Port Specific	PORTCFG[2:0]	Port Configuration: They are used to determine how 16 lanes are distributed among ports.	1. By default, the upstream port is port 0 2. PORTCFG[2:0] as 001b, 2Port-16Lane Configuration. 3. PORTCFG[2:0] as 010b, 3Port-16Lane Configuration. 4. PORTCFG[2:0] as 011b, 4Port-16Lane Configuration. 5. PORTCFG[2:0] as 100b, 5Port-16Lane Configuration. 6. PORTCFG[2:0] as 101b, 8Port-16Lane Configuration. These pins require the external 5.1K-ohm pull-up resistors or 330-ohm pull-down resistors.	
Power Supply and Signal Ground	VDDC	0.95V power. Used as digital core power pins.	1. Decoupling/Filtering capacitors must be added. 2. HI1812V101R-10 bead part is required at 0.95V power rail which VDDC/C_VDDC/VP.	□
	VDDR	1.8V power. Used as digital I/O power pins.	1. Decoupling/Filtering capacitors must be added. 2. HI1812V101R-10 bead part is required at 0.95V power rail which VDDC/C_VDDC/VP.	□
	C_VDDR	1.8V power. Used as reference clock power pins.		□
	C_VDDC	0.95V power. Used as reference clock power pins.		□
	VPH	1.8V power. Used as PCI Express analog high voltage power pins.		□
	VP	0.95V power. Used as PCI Express analog and core power pins.		□
	VSS	Used as ground pins.		□

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3.5. For GENIII PI7C9X3G1224GP

Section	Remark	Done
Block Diagram	● Customer should provide the block diagram to check	□
Port Configuration	● What port configuration in customer`s circuit.	□
Internal / External Clock Design In Circuit	● What kind of the clock source in customer`s circuit.	□
Reference Clock Source Type	● What kind of the clock type in customer`s circuit.	□
Parallel / Serial Hot Plug function implement (if chip supported)	● Does customer implement the hot plug function in circuit?	□
PM L1.1 function implement (if chip supported)	● Does customer implement the PM L1.1 function in circuit?	□
Fan out mode(Normal operate) or CDEP(Cross-Domain End-Device) Mode	● What kind of operation mode in customer`s circuit.	□

Section	Signal Name	Check Item	Remark	Done
Clock Source	REFCLKP_0[2:0] REFCLKN_0[2:0] REFCLKP_1[2:0] REFCLKN_1[2:0]	Connect to external 100MHz differential clock	1. The reference clock input is an unterminated AC coupled input. So the off-chip clock source must be terminated with both serial and parallel resistor network. Please refer to PCIe CEM specification for detail on how to realize on-board termination implementation. 2. When connecting the wires, please pay attention and check if the polarity and netnames are correct between your wires and the pins on the other side.	□
Reference Clock Input Pairs	REFCLKIP_0 REFCLKIN_0 REFCLKIP_1 REFCLKIN_1	Connect to external 100MHz differential clocks for the integrated reference clock buffers in two groups	4. If the reference clock input is HCSL type, it can be DC coupled. 5. If the 100MHz clock source is not in HCSL protocol (ex: LVPECL, LVDS, etc), please refer to application note to add application circuit. 6. When connecting the wires, please pay attention and check if the polarity and netnames are correct between your wires and the pins on the other side.	□
Reference Clock Output Pairs	REFCLKOP_0[7:0] REFCLKON_0[7:0] REFCLKOP_1 [5,4,1,0] REFCLKON_1 [5,4,1,0]	100MHz external differential HCSL clock outputs from integrated reference clock buffer in six groups	1. Group 0 is REFCLKOP_0[7:0] and REFCLKON_0[7:0] 2. Group 1 is REFCLKOP_1 [5,4,1,0] and REFCLKON_1 [5,4,1,0] 3. Group 0 has 8 fanouts for all of the ports in one tile 4. Group 1 has 4 fanouts for all of the ports in one tile 5. When connecting the wires, please pay attention and check if the	□

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Section	Signal Name	Check Item	Remark	Done
			polarity and netnames are correct between your wires and the pins on the other side.	
Reference Clock Power Down	CKBUFPD0_L CKBUFPD1_L	Integrated Reference Clock Buffer Power Down Signals: These signals CKBUFPDx_L, where x=0 and 1, are used to shut down the corresponding clock buffer in tile x. When CKBUFPDx_L is asserted low, the integrated reference clock buffer and reference clock outputs are disabled.		
Transmit and Receive Differential Pairs	PETP_0[15:0] PETN_0[15:0] PETP_1[11:8;3:0] PETN_1[11:8;3:0] PERP_0[15:0] PERN_0[15:0] PERP_1[11:8;3:0] PERN_1[11:8;3:0]	AC coupling capacitors of 220 nF should be placed at the same location (as close as possible) and should not be staggered from one trace to the other within the pair	4. This applies to all differential pairs of each port 5. It is fine to open un-used lane. 6. When connecting the wires, please pay attention and check if the polarity and netnames are correct between your wires and the pins on the other side.	□
RESET	PERST0_L PERST1_L	Pull-up and connect to the system reset		□
External Reference Resistor	RESREF_0 RESREF_1	Each tile has its own reference resistor connection signal. Attach RESREF_x, where x=0 and 1, an external resistor with the precision of 200 ohm 1% 100-ppm/C to ground on the board. The reference resistor is used for calibration of RX and TX termination when the chip comes out of reset or a manual PHY tuning request is made		□
EEPROM	EECK EEDI EEDO EECS_L	SPI Bus Serial EEPROM	4. Suggest reserving eeprom circuit for safety. 5. AT25XE011 or compatible SPI EEPROM parts. 6. EECK frequency is 1.58M. Please make sure selected SPI eeprom can support.	□
JTAG	TCK TMS TDO_[1:0] TDI TRST_L JTAG_SEL_L	TMS and TRST_L Signals should be pulled low through a 330 ohm pull-down resistor if JTAG boundary scan function is not implemented JTAG_SEL_L for switch JTAG and PHY CR parallel	3. Pull-down through 330 ohm at TMS and TRST_L for normal operation. 4. Pull-up through 5.1K ohm at JTAG_SEL_L for normal operation	□

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Section	Signal Name	Check Item	Remark	Done
		register interface control.		
Serial Hot-Plug	SHCL_I2C_0 SHCL_I2C_1	It is connected to SCL pin of all I2C IO expanders	3. I/O expander configure serial hot-plug port as follow Example 1. Set 011 at A[2:0] of I/O expander when SHP port configured as Port 4 of 3G1632GP. Example 2. Set 100 at A[2:0] of I/O expander when SHP port configured as Port 5 of 3G1632GP. 4. Plug high this pin through 5.1K ohm resistor if do not implement serial hot plug function.	□
	SHDA_I2C_0 SHDA_I2C_1	It is connected to SDA pin of all I2C IO expanders		□
	SHPCINT0_L SHPCINT1_L	It is connected to INT# output pin of all I2C IO expanders		□
	GPIO0[31]/SURPRISE_HP_0	Disable Surprise Hot Plug Function: This signal is used to disable surprise hot function if SURPRISE_HP=0.	3. Pull-down through 330 ohm at SURPRISE_HP_0 for operational type is “managed”. 4. Pull-up through 5.1K ohm at SURPRISE_HP_0 for operational type is “surprised”.	
SMBUS /I2C	SDA_I2C	Bi-Directional System management or I2C Bus Data. This pin requires an external pull-up resistor.	NC if do not implement SMBUS/I2C function	□
	SCL_I2C	Bi-Directional System management or I2C Bus Clock. This pin requires an external pull-up resistor.	NC if do not implement SMBUS/I2C function	
	I2C_ADDR[2:0]	These pins are used to configure the value of the three least significant bits of the switch 7-bit Slave address. These pins require the external 5.1K-ohm pull-up resistors or 330-ohm pull-down resistors.	These pins require the external 5.1K ohm pull-up resistors or 330 ohm pull-down resistors.	□
	EECK/ Debug_Mode_EN_L	EEPROM Clock: Clock signal to 4-wire EEPROM interface. Debug Mode Enable (Debug_Mode_EN_L): During system initialization, EECK acts as the Debug_Mode_EN_L pin. In debug mode, it need be tired to low through a 330-ohm pull-down resistor. This pin has internal pull-up resistor. If no board trace is connected to this pin, the	Pull-up through 5.1K ohm at EECK/ Debug_Mode_EN_L for normal operation	□

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Section	Signal Name	Check Item	Remark	Done
		internal pull-up resistor of this pin is enough. However, if pin is connected to a board trace and not driven, it is recommended that an external 5.1K-ohm pull-up resistor be used.		
Misc Signals	GPIO[30]/ PHY_SRAM_ BYPASS	PHY_SRAM_BYPASS: when set, it bypasses PHY SRAM. This pin has internal pull-up. If no board trace is connected to this pin, the internal pull-up resistor of this pin is enough. However, if pin is connected to a board trace and not driven, it is recommended that an external 5.1K- ohm pull-up resistor be used.	Pull-up through 5.1K ohm at GPIO[30]/ PHY_SRAM_BYPASS for normal operation	□
	FATAL_ERR0_L/HOT_PLUG_EN0_L FATAL_ERR1_L/HOT_PLUG_EN1_L	Hot Plug Function Enable: Each switch tile has its own hot plug function enable signal to determine the downstream port is capable of handling either managed or surprised hot plug events. Besides, the GPIO pins would be redefined for hot-plug function if HOT_PLUG_ENx_L = 0. In total, there are 2 switch tiles and the corresponding hot plug enable signals are HOT_PLUG_EN_x, where x=0 and 1.	3. Pull-up through 5.1K ohm at this pin for disable hot plug function 4. Pull-down through 330 ohm at this pin for enable hot plug function	□
	INTA0_L/PM_L11_EN0_L INTA1_L	PM L1.1 Function Enable: Each switch tile has its own PM L1.1 function enable signal to determine the downstream port is capable of PM L1.1 function. Besides, the GPIO_x [15:8] pins would be redefined for PM L1.1 function if PM_L11_ENx_L = 0. In total, there are 2 switch tiles and the corresponding PM L1.1 function enable signals are PM_L11_ENx_L, where x=0 and 1.	3. Pull-up through 5.1K ohm at this pin for disable PM L1.1 function 4. Pull-down through 330 ohm at this pin for enable PM L1.1 function	□
	CLKBUF_CMOS_EN0_L CLKBUF_CMOS_E	Intneral Reference Clock Input Select: Used to select the input for internal	Pull-up through 5.1K ohm at both CLKBUF_CMOS_EN0_L and CLKBUF_CMOS_EN1_L for normal	□

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Section	Signal Name	Check Item	Remark	Done
	N1_L	reference clock buffer is came from HCSL or CMOS. When CLKBUF_CMOS_ENx_L is asserted, CMOS is selected.	operation	
	CHIPMODE[1:0]	Chip Operational Mode: These two input signals decide at which operational mode the chip is chosen	Pull-down through 330 ohm at CHIPMODE[1:0] for normal operation	□
	CKMODE_0 CKMODE_1	Clock Operational Mode: Each tile has its own clock operation mode signal (CKMODE_0 for tile #0 while CKMODE_1 for tile #1), which defines the relationship between physical SERDES lanes and reference clock. When CKMODE is 0, all 16 lanes (0 ~15) of SERDES are driven by one reference clock source. When CKMODE is 1, 16 lanes of SERDES are driven by two separate reference clock source via three pairs of reference clock inputs.	3. Pull-up through 5.1K ohm at this pin for CKMODE=1 function, which REFCLKP/N [3:0] pairs need to connect from external clock source. 4. Pull-down through 330 ohm at this pin for CKMODE=0 function, which REFCLKP/N [3:1] pairs are unused, and required pull-down 330 ohm at REFCLKP/N [3:1] pin. External clock source only connect to REFCLKP/N [0] pair.	□
	SWP_MODE	Switch Partition Mode: The switch can be partitioned into multiple switches based upon the value of SWP_MODE	Pull-down SWP_MODE pin for normal operate.	□
	PDC0_L[7:0] PDC1_L[5,4,1,0]	Present Detect: When PDC_L is asserted low, it indicates this port is present. Otherwise, it indicates this port is absent.	5. Pull-down if chip to chip connection. 6. Pull-up and connect to pin PRSNT of slot if chip to slot connection. 7. NC if unused. 8. When connecting the wires, please pay attention and check if the netnames are correct between your wires and the pins on the other side.	□
Port Specific	PORTCFG_0[2:0] PORTCFG_1[2:0]	Port Configuration: Each switch tile has its own port configuration, which determines how 16 lanes are distributed among ports. In total, there are two switch tiles and the corresponding port configuration signals are PORTCFG_x[2:0], where x=0 and 1.	1. By default, the upstream port is port 0 These pins require the external 5.1K-ohm pull-up resistors or 330-ohm pull-down resistors.	

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Section	Signal Name	Check Item	Remark	Done
Power Supply and Signal Ground	VDDC	0.95V power. Used as digital core power pins.	3. Decoupling/Filtering capacitors must be added. 4. HI1812V101R-10 bead part is required at 0.95V power rail which VDDC/C_VDDC/VP.	□
	VDDR	1.8V power. Used as digital I/O power pins.	3. Decoupling/Filtering capacitors must be added. 4. HI1812V101R-10 bead part is required at 0.95V power rail which VDDC/C_VDDC/VP.	□
	C_VDDR	1.8V power. Used as reference clock power pins.		□
	C_VDDC	0.95V power. Used as reference clock power pins.		□
	VPH	1.8V power. Used as PCI Express analog high voltage power pins.		□
	VP	0.95V power. Used as PCI Express analog and core power pins.		□
	VSS	Used as ground pins.		□

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